Binary Trees

specific kind of tree with every node having 2 children

just 2 children.

left right

→ makes the structure simpler than a generic tree
→ extremely powerful for various computational purposes

Real life example.

→ Family tree with 2 parent
→ Yes/No Decision making
→ Tournament Bracket

Industry example

→ Binary DT in AI
→ Database Searching (mainly BST)
→ Binary Trees in compilers

Binary Tree node

Now no list is needed, just a left & right will do

2 subtree



Print function

→ now loose code would be required

also now no loop now & only left, right would do

inorder preorder post order

No duplicate

Tree from inorder and preorder

not to make mistake of in S[0]

Diameter of a binary tree

max^m distance b/w 2 nodes

→ Check balanced → Sum of nodes → find min max

→ Tree from inorder and post order

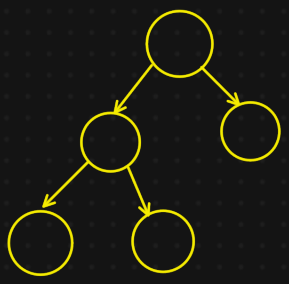
→ Tree from inorder and pre-order

Types of binary tree

1] On basis of children

i) Full binary tree

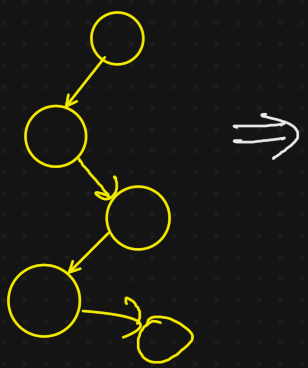
a binary tree in which each node either has 0 or 2 children



2) Degenerate tree

A tree in which each internal node has one child.

≈ equal to LL



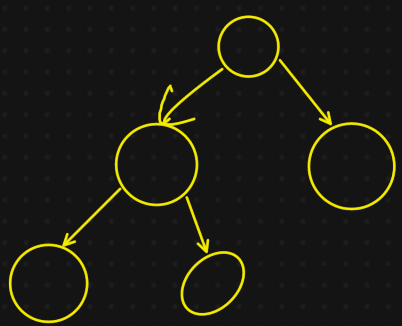
Skewed (one child either on left or on right)



2) Division Based on Completion of level

1) Complete Binary tree (CBT)

All the level are completely filled except possibly the last level and the last level has keys as left as possible.



need not to be full

if last level has 1 node then it should be left one.

2. Perject Binary Tree

A perfect BT is a tree in which all the internal nodes have 2 children and all leaf nodes at the same level.

eg. ancestors of a child



no. of leaf nodes = (internal nodes) + 1

no. of nodes $= 2^{n+1} - 1$

3. Balanced BT

type of BT in which difference b/w the height of left and right subtree for each node is 0 or 1.

Operation on Binary Tree

If we have to insert in a binary tree, how can we do that?

→ No major logic or something I can use

→ how to traverse?

have some special kind of trees in this.

Binary Search Tree

AVL tree

Red Black etc.